



On a new species, *Cathetocephalus leucas* (Tetraphyllidea: Cathetocephalidae) from the bull shark, *Carcharhinus leucas* (Valenciennes, 1839) from Bay of Bengal, Visakhapatnam coast, Andhra Pradesh, India.

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ABSTRACT. A new and rare species of *Cathetocephalus* was collected from the spiral valve of bull shark, *Carcharhinus leucas* (Valenciennes, 1839) at Visakhapatnam coast. Four parasites were obtained from the host, and were characterized by the presence of characteristic and unique v-shaped scolex with the two arms drawn into thin, narrow and slender ribbon-like structures at the tips. This feature of scolex separates these parasites from the already reported species of the genus, *C. thatcheri* Dailey and Overstreet, 1973; *C. australis* Schmidt and Beveridge, 1990 and *C. limbatus* Pramanik and Manna, 2006. The present specie is named as *Cathetocephalus leucas*.

Keywords: *Carcharhinus leucas*, *Cathetocephalus leucas*, scolex, spiral valve

INTRODUCTION

Sharks among elasmobranchs constitute an important group of fishes in east coast of India due to their abundant occurrence, cosmopolitan distribution and high marketable value. They have an escalating pace of demand due to high human consumption rate, medicinal value and decorative purposes. They serve as excellent hosts for a range of host-specific parasites, which can be used as 'biological indicators'. Sharks offer an exceptional habitat to a wide spectrum of cestode parasitic fauna, in particular diphyllids, tetraphyllids, lecanicephalids and trypanorhynchids. Literature review supported the fact that fairly, a good amount of work on cestode parasites of elasmobranchs has been contributed from all over the

world but taxonomic work on cestodes of sharks is infrequent and is contributed by Linton (1921, 1924), Sproston (1948), Young (1954), Dailey and Overstreet (1973), Rego and Mayer (1976), Rego (1977), Cheung *et al.*, (1981), Shinde *et al.*, (1984), Sarada *et al.*, (1984, 1986), Caira (1985, 1986, 1990, 1992), Dailey and Vogelbein (1990), Nock and Caira (1988), Schmidt and Beveridge (1990), Caira and Ruhnke (1990), Caira and Gavarrino (1990), Caira and Runkle (1993), Cislo and Caira (1993), Ruhnke (1996), Vijayalakshmi *et al.*, (1996), Caira *et al.*, (1997), McKenzie and Caira (1998), Scholz *et al.*, (1998), Brooks *et al.*, (1999) and Caira *et al.*, (1999). The comprehensive study on the systematics of cestode parasites of sharks from this coast is lagging behind and not yet fully exposed and such studies on cestodes of sharks from this region will enable taxonomists to disclose many new taxa, which are still awaiting discovery. In the present study, a new and a

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rare species, *Cathetocephalus leucas* sp. nov is described from the spiral valves of *Carcharhinus leucas*. The identification of the genus was done by following the standard books like Yamaguti (1959), Schmidt (1986), Khalil *et al.*, (1994) and recent literature pertaining to the genus (Dailey and Overstreet, 1973; Schmidt and Beveridge, 1990; Pramanik and Manna, 2006)

MATERIAL AND METHODS

About 125 *Carcharhinus leucas* (Valenciennes, 1839) were examined thoroughly for the cestode parasites in a survey to study cestode parasitic fauna from the sharks at Visakhapatnam coast, Bay of Bengal. Spiral valves of the intestine offer an excellent habitat to cestode parasites. Spiral valves were separated into physiological saline to remove excess mucus and were cut open with a longitudinal incision. Parasites were carefully isolated from the spiral valves and collected in petridishes filled with saline solution. Due care was taken to prevent the scolex as well as strobila of parasites from being damaged as scolex being the key feature for the identification of a cestode. Parasites were flattened between two slides or under the pressure of slide and a coverslip, post fixed in A.F.A (alcohol, 85 ml; formalin, 10 ml and acetic acid, 5 ml) and stained with alum carmine. Permanent whole mount preparations were made by employing the

conventional techniques. Figures were drawn with the aid of camera lucida. Measurements are given in millimeters, unless otherwise stated. A new and a rare cestode, *Cathetocephalus leucas* sp. nov was encountered during the study from the spiral valve of the intestine.

RESULTS

Class: Cestoda

Sub-class: Eucestoda Southwell, 1925

Order: Tetraphyllidea Carus, 1863

Family: Cathetocephalidae Dailey and Overstreet, 1973

Genus: *Cathetocephalus* Dailey and Overstreet, 1973

Cathetocephalus leucas n. sp.

Only four specimens were obtained from the spiral valve of a single bull shark, *Carcharhinus leucas* (Valenciennes, 1839) and were found to be deeply embedded into the intestinal wall.

DESCRIPTION (Based on measurement of four parasites, Plate-I, Figs.1-2): Parasites long, thin, creamy white and measure 7–8 cm in length. Worms slightly acraspedote and apolytic in nature. Scolex

Cathetocephalus leucas n.sp

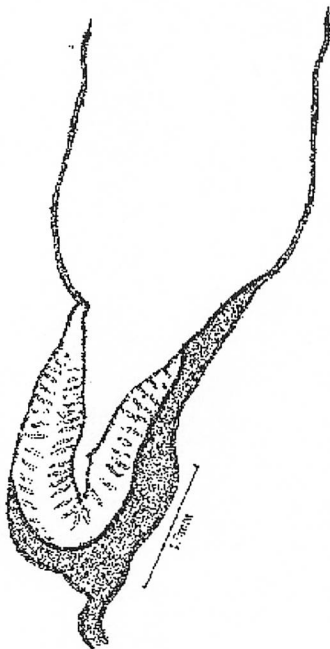


Fig. 1. Scolex

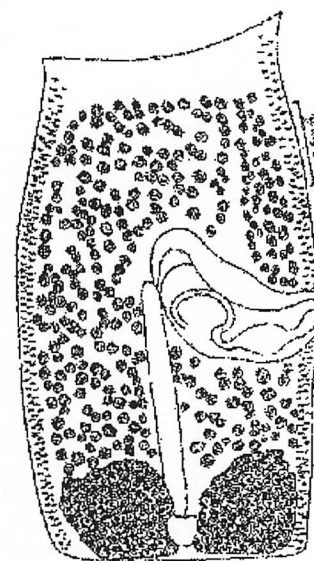


Fig. 2. Mature Proglottid

fleshy, with a rugose dorsal surface and a smooth ventral region. Dorsal surface of scolex bears two parallel rows of fleshy papilliform proliferations on the margins of upper surface of the scolex. Scolex perpendicular with a v-shaped bent in the centre and the two arms being drawn into thin, long, narrow and slender ribbon like structures. These ribbon like structures are thin and delicate and are difficult for separation from tissues during collection. Accessory hooks or suckers not seen. Each arm of the scolex measures 2.50–2.63 from the base and the ribbon measures 2.9–3 long. Neck thick, long and measures 5–6.57 x 0.16–0.18. Strobila comprises 55–110 proglottids. Proglottids 3–4 times longer than broad. Immature proglottids measure 0.58–1.21 x 0.29–0.47. Mature region starts from 0.26–0.83 behind the neck. Mature proglottids measure 2.05–4.08 x 0.32–0.58. Gravid proglottids not obtained. Testes small, 148–280 in number, spherical measuring 0.03–0.08 in diameter. Testes pre-ovarian, filling the full length in medullary region, but not reaching upto the anterior margin of the proglottid. A space is left in the anterior most region of the proglottid characteristically. Cirrus sac large, pear-shaped and measures 0.11–0.34 x 0.20–0.55. Cirrus unarmed measuring 0.08–0.10 in length. Genital pores irregularly alternate and open slightly posterior to the mid margin of the proglottid. Vagina opens anterior to the cirrus and emptying into common genital atrium. Ovary large, with two wing shaped lobes, posterior in position. Each lobe measures 0.2–0.72 x 0.11–0.42. Vitellaria follicular, lateral in position extending in the entire margin of the proglottid in cortical region. Uterus tubular, originates from ovarian isthmus and extends anteriorly.

Host: *Carcharhinus leucas* Valenciennes, 1839

Habitat: Spiral valve of the intestine

Location: Visakhapatnam coast, Andhra Pradesh, India

No. of specimens: Four

Deposition of specimens: Holotype and Paratypes deposited in the Parasitology Laboratory, Department of Zoology, Andhra University, Visakhapatnam.

Etymology: Named after the species name of the host fish.

DISCUSSION

Cathetocephalidae is a small family of the order Tetraphyllidea with *Cathetocephalus* as the only

genus of the family. Thatcher (1961) erected this genus with *C. thatcheri* as its type-species from *Carcharhinus limbatus* Val, 1840 in the coastal waters of Mississippi and Louisiana. The genus is characterized by the presence of the perpendicular head and hence the name is derived from greek words “Katheto” and “Kephalos”. At that time, Thatcher considered it as *Pillersium owenium* Southwell, 1927 because of T-shaped scolex but Dailey and Overstreet (1973) re-examined Thatcher's original material and named it as *Cathetocephalus thatcheri*, which was the type and the only species in the genus and erected the family Cathetocephalidae for it. However, in his description, he failed to record the number of testes. Later, Watson and Thorson (1976) redescribed this species from Lake Nicaragua and recorded the number of testes to be 300–400. Schmidt (1986) also recognized a fourth family, the Cathetocephalidae Dailey and Overstreet, 1973, within the Tetraphyllidea in addition to the Onchobothriidae, Phyllobothriidae and Triloculariidae. Euzet (1994), in a most recent and comprehensive treatment of the group, recognised eight families of tetraphyllideans of which Cathetocephalidae was one among them.

Schmidt and Beveridge (1990) after a long span of time described a second species, *C. australis* of the family Cathetocephalidae from the spiral intestine of *C. amblyrhynchoides* from Australian waters. Small scolex, craspedote proglottids, small ovarian lobes, less number of testes and follicular vitellaria, characterize this species. Pramanik and Manna (2006) described *C. limbatus* from the spiral valves of *Carcharhinus limbatus* from India, which is characterized in having scolex with four suckers, oval cirrus pouch, U-shaped ovary and follicular vitellaria.

The present species shows some resemblances and differences with these three species. It comes closer to *C. thatcheri* in not having suckers but differs in many characters like number of testes, number of proglottids, craspedote strobila and shape of scolex. It resembles *C. australis* in having follicular vitellaria and number of testes but differs in the shape of scolex, number of proglottids, position of genital opening, and nature of strobila. It comes closer to *C. limbatus* in having follicular vitellaria, acraspedote strobila but differs in the shape of scolex and number of testes. The present parasites characteristically differ from all these three species in having an unique scolex with an upward bent in the centre, giving v-shape to the scolex with its two arms drawn into a narrow, slender ribbon

Table I. Comparison of present new species of *Cathetocephalus* with other related species of the genus

	<i>C.thatcheri</i> Dailey et Overstreet, 1973	<i>C.australis</i> Schmidt & Beveridge, 1990	<i>C.limbatus</i> Pramanik & Manna,2006	Present new species
1. Host	<i>Carcharhinus leucas</i> (Val, 1839)	<i>C.amblyrhynchoides</i> (Whitley, 1934)	<i>Carcharhinus limbatus</i> (Val, 1840)	<i>Carcharhinus leucas</i> (Val, 1839)
2. Size	10.1-10.5 cm	5 cm	4.80 cm	7-8 cm
3. No. of proglottids	228-325, Slightly craspedote	Slightly craspedote	94-98, slightly acraspedote	55-110, Slightly acraspedote
4. Size of scolex	4-10	1.12-2.9	1.7 with four suckers	2.5-2.63 x 0.5-0.7, suckers not seen
5. Width at broader Surface	0.55-1.5	0.5-0.96	-	
6. Neck	0.55-1.5	0.21-0.44	2.26 x 0.3	5.0-6.57 x 0.16-0.18
7. Mature proglottids	2.9-4.1 x 0.86-1.21	0.92-1.68 x 0.77-1.60	1.71 x 0.54	2.05-4.08 x 0.32-0.58
8. Gravid proglottids	5.5 x 1.0	1.68-2.98 x 1.38-1.76	-	-
9. Testes	300-500, not reaching anterior margin of proglottid	125-205, occupies anterior margin of proglottid	132-134, reaching upto anterior margin of proglottid	148-280, not reaching upto anterior margin of proglottid.
10. Cirrus sac	0.53-0.59 x 0.25-0.34	0.30-0.63 x 0.16-0.32	0.3 x 0.2	0.11-0.34 x 0.20-0.55
11. Genital opening	Lateral, irregularly alternate, slightly posterior mid region of proglottid	Lateral, irregularly alternate, mid region of proglottid	mid-portion of segment	lateral, irregularly alternate, slightly posterior to mid region of proglottid
12. Ovary	dumb-bell shaped 0.99-1.13 x 0.34-0.50	0.20-0.40 x 0.40-0.84	U-shaped, 0.45 x 0.2	Winged, 0.2-0.72 x 0.11-0.42
13. Vitellaria	Granular, not extending entire length of proglottid	Follicular, forming a sleeve around vitellaria	Follicular, 2-3 rows	Follicular

* Note: Measurements are given in millimeters, unless otherwise stated.

like structures at tips. A table comparing the present parasites from the already described species is given in Table-I. Based on these differentiating characters, it is felt justified erecting it to the status of a new species.

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